



PARENTAL MONITORING APP SYSTEM

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Section 1. Project Overview

1.1 Project Overview

The Parental App Monitoring System is a software application designed to help parents monitor and control their children's smartphone usage. The system allows parents to track applications, monitor screen time, and receive activity reports.

It also provides features such as setting usage limits and blocking inappropriate apps to ensure a safe digital environment for children.

The main goal of this project is to improve children's digital safety and help parents manage device usage effectively. The system will be developed as a simple and user-friendly prototype that demonstrates system analysis and design concepts.

1.2 Project Scope

Project Includes (In Scope): "Internal scope"
• User registration and login for parents and children
• Monitoring installed applications
• Tracking screen time and app usage
• Setting application time limits
• Blocking inappropriate applications
• Storing activity data and reports
• Managing parent and child accounts
Project Includes (In Scope): "External scope"
• Sending notifications to parents
• Generating and sending activity reports
• Accessing child device permissions
• Synchronizing data between parent and child devices
• Interaction with mobile operating system services

Project Excludes (Out of Scope):

- Hardware development
- Supporting old devices
- Working without internet
- Monitoring calls or messages
- Full device control

1.3 Assumptions

- The child has a smartphone.
- The app is installed on the child's device.
- Parents have access to the app.
- Internet connection is available.
- The child does not remove the app.
- The device supports the application

1.4 Constraints

- The system requires an internet connection.
- Performance depends on device type and operating system.
- Battery life may affect tracking and performance.
- Some features require special permissions.
- Privacy laws may limit some monitoring features.
- The system may not work properly on outdated devices.

Section 2. Project Start-up

2.1 Project Life Cycle

In this project (Parental Control System), the life cycle explains how the system is planned, analyzed, designed, developed, tested, and deployed to help parents monitor and manage their children's mobile usage in a safe and organized way.

Planning	1. Define the project idea (Parental Control System) 2. Identify user needs (parents and children) 3. Set project goals 4. Determine basic system requirements
Analysis	1. Study how parents monitor children's mobile usage 2. Identify required features (tracking, blocking, time limits) 3. Analyze system requirements 4. Understand user expectations
Design	1. Design the user interface (UI) 2. Define system structure and workflow 3. Design application screens 4. Plan how the system components interact
Implementation	1. Develop the application (coding) 2. Implement app tracking features 3. Add screen time control 4. Add blocking and reporting features 5. Connect system components

2.2 Methods, Tools, and Techniques

Methods:

- Incremental development (building the system step by step)
- Agile methodology for flexible development

Tools:

- Mobile application development tools
- Programming languages (such as Java or Kotlin / Flutter)
- Database systems (for storing user data)
- GPS and mobile tracking APIs

Techniques:

- User interface (UI) design techniques
- Data tracking and monitoring techniques
- Notification system techniques
- Data analysis and reporting

2.3 Estimation Methods and Estimates

Project Estimates:

Type	Estimates
Time Estimation	3 – 6 weeks
Cost Estimation	Low to medium depending on tools and services used
Effort Estimation	Medium effort due to multiple system features
Resource Estimation	Developer, tester, analyst, and software tools

Effort Estimation:

- The project requires a medium level of effort.
- Effort is needed for developing features like tracking, blocking, and reporting.
- Additional effort is required for testing and fixing errors.
- Collaboration between development and testing may be needed.
- Continuous updates and improvements increase the effort slightly.

Time Estimation:

Phase	Description	Duration
Planning & Analysis	Understanding the problem, defining scope, and identifying system requirements.	3 – 5 days
System Design	Designing system architecture, database structure, and UI/UX layout.	4 – 7 days
Implementation (Coding)	Developing core features such as tracking, monitoring, and reporting modules.	14 – 21 days
Testing	Testing system functionality, fixing bugs, and ensuring performance quality.	4 – 6 days
Deployment	Installing and releasing the system for use on target devices.	1 – 2 days
Maintenance	Continuous updates, bug fixes, and system improvements after release.	Ongoing

- **Total Estimated Effort: approximately 22–35 working days across all project phases**
- **Total Estimated Schedule: The project is expected to be completed within 3–6 weeks**

2.4 System Requirements

The system requirements define what the system must do and the constraints under which it must operate, ensuring a clear guide for the design and development phases.

Functional Requirements
The system shall allow parents to monitor their children’s device activities.
The system shall track the child’s real-time location using GPS.
The system shall allow parents to block specific applications.
The system shall send notifications and alerts to parents.
The system shall provide usage reports such as screen time and app usage.
The system shall allow parents to set usage limits on devices
The system shall allow viewing of activity history.

Non-Functional Requirements
The system shall ensure data security and user privacy.
The system shall be user-friendly and easy to navigate.
The system shall respond within a short time to user requests (good performance).
The system shall be compatible with Android devices.
The system shall operate reliably with minimal downtime.
The system shall handle multiple users efficiently

Hardware and Software Requirements
Smartphones (Android devices for parents and children).
Internet connection for real-time monitoring.
GPS service enabled on devices.
Backend server for data storage and processing.
Mobile development tools such as Android Studio or Flutter.
Database system such as Firebase or SQL Server.

2.5 Risk Management

Risks and Mitigation

- Technical Risk:** The system may not work properly on all devices.
Mitigation: Test the application on different devices and Android versions.
- GPS Accuracy Risk:** Location tracking may be inaccurate at times.
Mitigation: Use reliable GPS services and optimize updates.
- Security Risk:** Unauthorized access to user data.
Mitigation: Apply authentication and data encryption.
- User Permission Risk:** Users may deny required permissions.
Mitigation: Provide clear instructions and simple permission requests.
- Performance Risk:** App performance may slow due to background services.
Mitigation: Optimize code and reduce unnecessary processes.

2.6 Schedule Allocation:

Activities with Effort

Phase 1: Planning & Analysis (3–6 days)

Activities	Effort
Define project problem and scope	1 day
Gather requirements	1–2 days
Analyze requirements	2 days
Confirm feasibility & planning	1 day

Phase 2: System Design (4–7 days)

Activities	Effort
Design system architecture	2 days
Design database	1–2 days
Design user interface	2–3 days

Phase 3: Implementation (2–3 weeks)

Activities	Effort
Develop core features (tracking, blocking, reporting)	7–10 days
Implement alerts & notifications	3–5 days

Phase 4: Testing (4–6 days)

Activities	Effort
Functional testing	2–3 days
Bug fixing	2–3 days

Phase 5: Deployment (1–2 days)

Activities	Effort
System installation	1 day
Final release & validation	1 day

Activities with Time : (Starting from 01/03/2026)

Activity	Start Date	End Date
Define project scope	01/03/2026	01/03/2026
Gather functional requirements	02/03/2026	02/03/2026
Gather non-functional requirements	03/03/2026	03/03/2026
Analyze system requirements	04/03/2026	04/03/2026
Feasibility study	05/03/2026	05/03/2026
Create system architecture	06/03/2026	07/03/2026
Design database schema	08/03/2026	09/03/2026
Design ERD and mapping	10/03/2026	10/03/2026
Design user interfaces (UI)	11/03/2026	13/03/2026
Develop parent module	14/03/2026	18/03/2026
Develop child monitoring module	19/03/2026	22/03/2026
Implement screen time tracking	23/03/2026	24/03/2026
Implement app blocking feature	25/03/2026	26/03/2026
Integration of system modules	27/03/2026	28/03/2026
Unit Testing	29/03/2026	30/03/2026
Integration Testing	31/03/2026	01/04/2026
System Testing	02/04/2026	02/04/2026
User Acceptance Testing (UAT)	03/04/2026	03/04/2026
Deployment and final setup	04/04/2026	04/04/2026

2.7 Budget Allocation:

Cost Item	Description	Estimated Cost
Development Tools & Software	IDEs, design tools, and development platforms	\$5,000
Server & Database Hosting	Cloud hosting, database storage, and backend services	\$12,000
APIs (GPS / Notifications / Maps)	External services for tracking and alerts integration	\$8,500
Testing Tools & Environment	Testing frameworks and device testing setup	\$7,000
Maintenance & Updates	Bug fixing, improvements, and system support	\$7,500
Security and Data Privacy	Implementation of encryption protocols and data protection standards.	\$10,000
Personnel Effort	Costs for developers, analysts, and testers throughout the project lifecycle.	\$35,000

Total Estimated Budget: \$85,000

The budget allocation ensures a balanced distribution of resources across all system components, supporting development, testing, deployment, and long-term maintenance of the system.

Section 3. Project Scheduling and Network Analysis

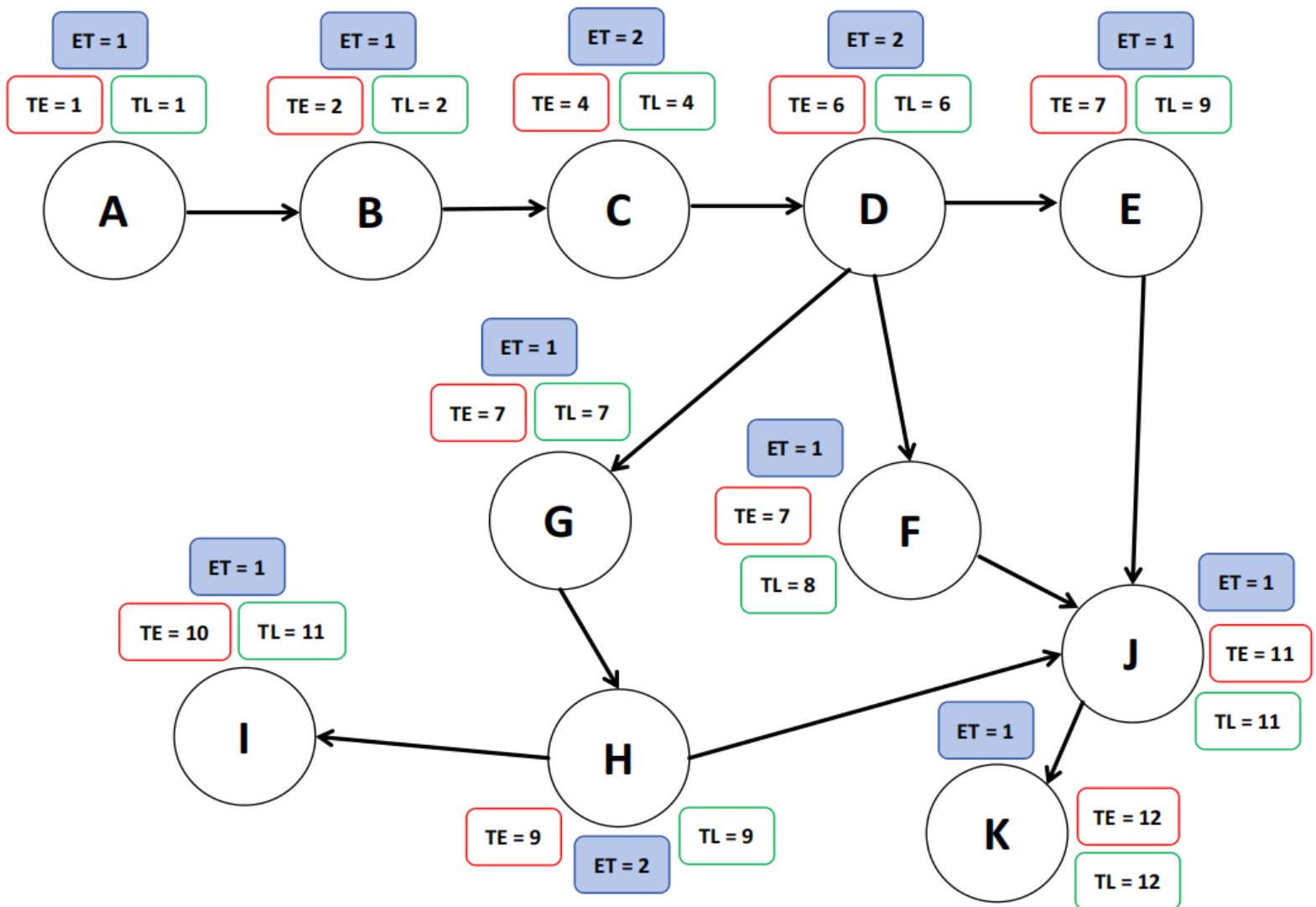
3.1 Work Breakdown Structure

Estimated				Actual				Dependence
ID	Task	Assigned to	Duration (Days)	Start Date	Finish Date	Start Date	Finish Date	
A	Define Project Scope	Farha Ibrahim	1	01/03	01/03	01/03	01/03	---
B	Identify objectives	Christina	1	01/03	02/03	01/03	02/03	A
C	Gather requirement	Christina	2	03/03	04/03	04/03	05/03	B
D	Analyze requirements	Fatma Ibrahim	2	05/03	06/03	06/03	07/03	C
E	Feasibility study	Caroline	1	07/03	07/03	08/03	08/03	D
F	Identify systems constraints	Fajr Adel	1	07/03	07/03	08/03	08/03	D
G	Draw Context Diagram	Caroline	1	07/03	07/03	08/03	08/03	D
H	Draw DFD	Fatma Mohamed	2	08/03	09/03	09/03	10/03	G
I	Draw Network Diagram	Fajr, Fatma Ibrahim	1	10/03	10/03	11/03	11/03	H
J	Prepare SRS document	Fajr Adel	2	10/03	11/03	11/03	12/03	E, F, H
K	Final approval	Fatma Mohamed	1	12/03	12/03	13/03	13/03	J

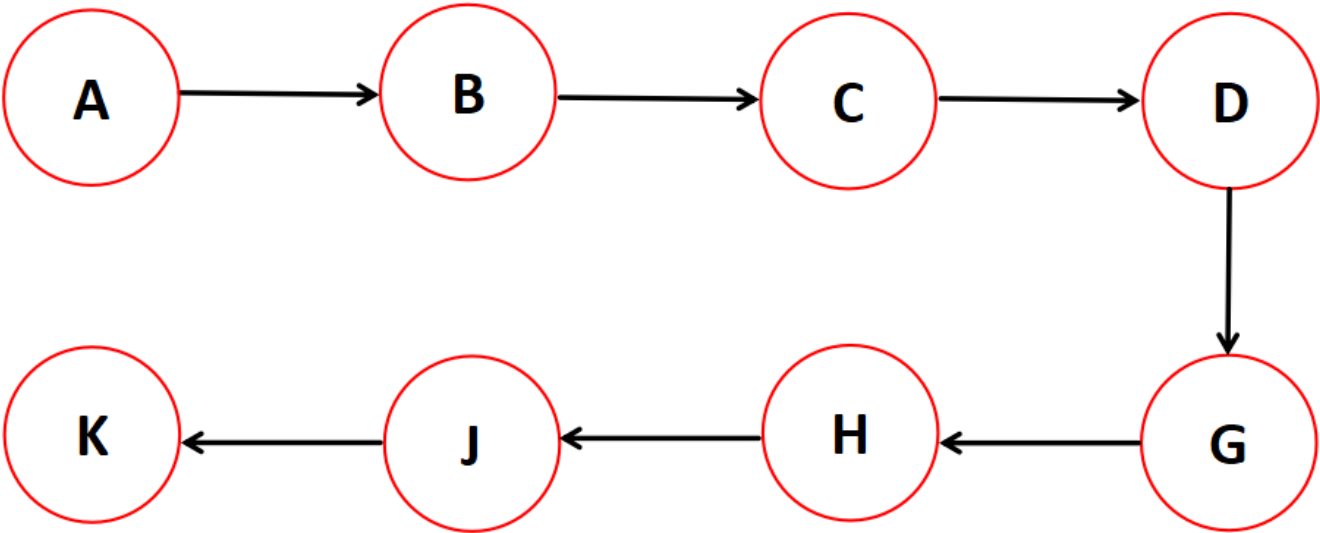
3.2 PERT:

Task	ET	TE	TL	Slack	Critical
A	1	1	1	0	Y
B	1	2	2	0	Y
C	2	4	4	0	Y
D	2	6	6	0	Y
E	1	7	9	2	N
F	1	7	9	2	N
G	1	7	7	0	Y
H	2	9	9	0	Y
I	1	10	11	1	N
J	2	11	11	0	Y
K	1	12	12	0	Y

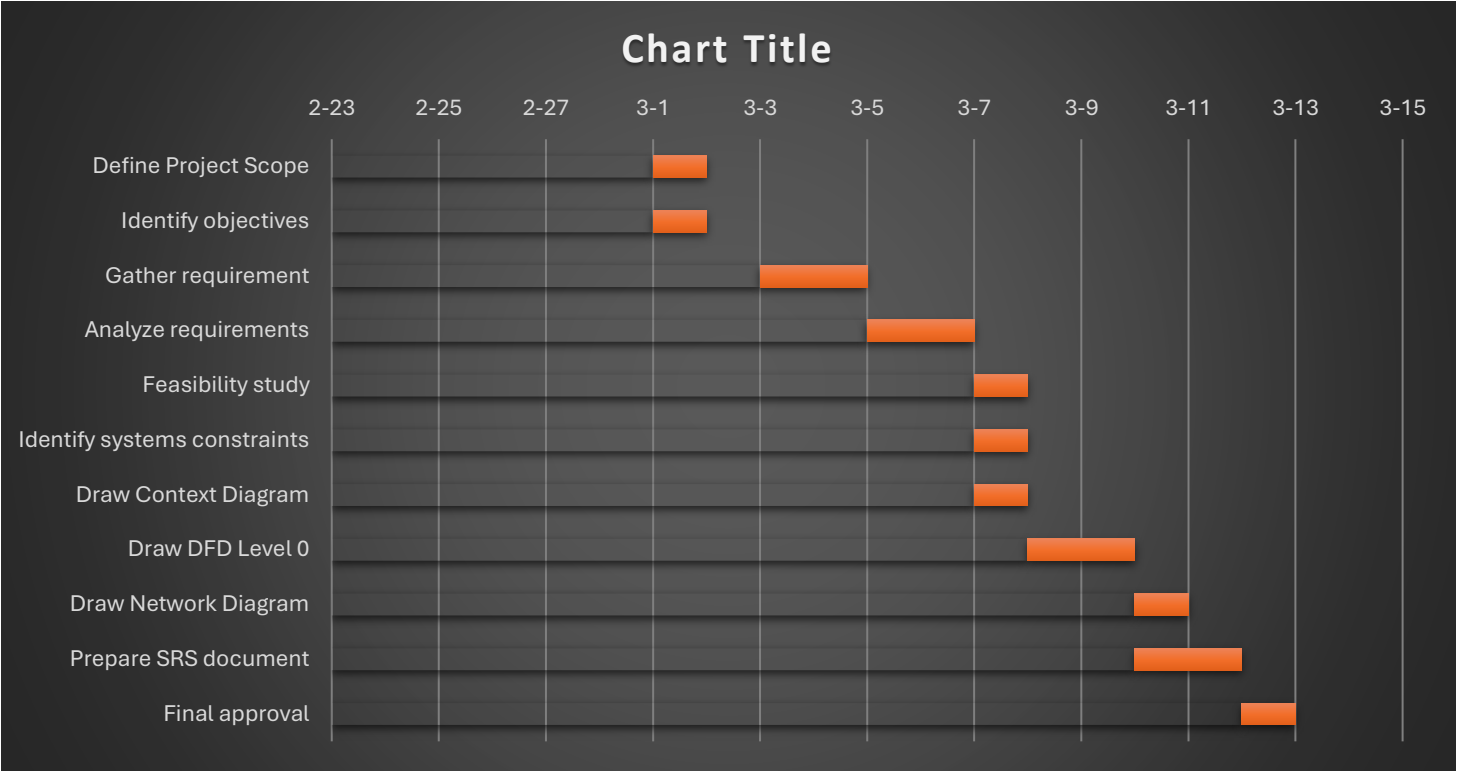
3.3 Network Diagram:



3.4 Critical Path



3.5 Gannt Chart



3.6 Feasibility Study:

One time costs :

Cost Item	Description	Cost
System Analysis	Designing the core system workflow and backend structure.	\$8,000
UI/UX Professional Design	Creating high-fidelity prototypes and user interfaces for parents and children.	\$5,000
Core Development	Full-stack programming for tracking, blocking, and reporting features.	\$25,000
Hardware & QA Lab Setup	Purchasing or leasing physical Android devices for rigorous testing.	\$10,000
Deployment & Store Setup	Google Play Console fee (\$25) and initial store optimization.	\$2,000
Total One-Time Costs		\$50,000

Recurring Costs :

Cost Item	Description	Annual Cost
Cloud Infrastructure & Hosting	Ongoing costs for servers, databases, and real-time data synchronization..	\$12,000
Premium API Subscriptions	Recurring fees for enterprise-level GPS tracking, Maps, and Notifications.	\$8,500
Security Audits & Compliance	Regular encryption updates and ensuring data privacy law compliance.	\$7,000
Technical Support & Maintenance	Fixing bugs, system improvements, and OS compatibility updates.	\$7,500
Total Recurring Costs		\$35,000

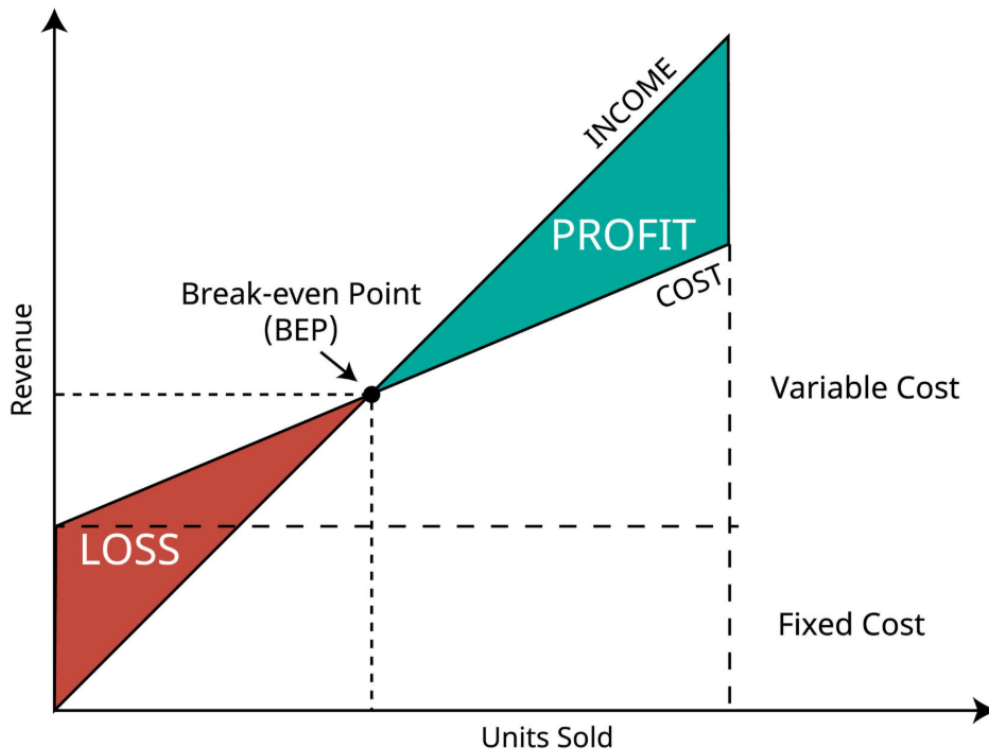
Economic :

- 1. Benefit per year = \$90,000**
- 2. Recurring cost per year = \$35,000**
- 3. Discount Rate = %12**
- 4. One-time cost = \$50,000**
- 5. 5 Years**

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Benefits	0	\$90,000	\$90,000	\$90,000	\$90,000	\$90,000	
PV Benefits	0	\$80,357	\$71,747	\$64,060	\$57,197	\$51,068	
NPV Benefits	0	\$80,357	\$152,104	\$216,164	\$273,361	\$324,429	\$324,429
Costs	\$50,000	\$35,000	\$35,000	\$35,000	\$35,000	\$35,000	
PV Costs	\$50,000	\$31,250	\$27,902	\$24,912	\$22,243	\$19,860	
NPV Costs	\$50,000	\$81,250	\$109,152	\$134,064	\$156,307	\$176,167	\$176,167
Yearly Cash flow	-\$50,000	\$49,107	\$43,845	\$39,148	\$34,954	\$31,208	
Overall Cash flow	-\$50,000	- \$893	\$42,952	\$82,100	\$117,054	\$148,262	
Overall NPV							\$148,262
Overall ROI							0.892

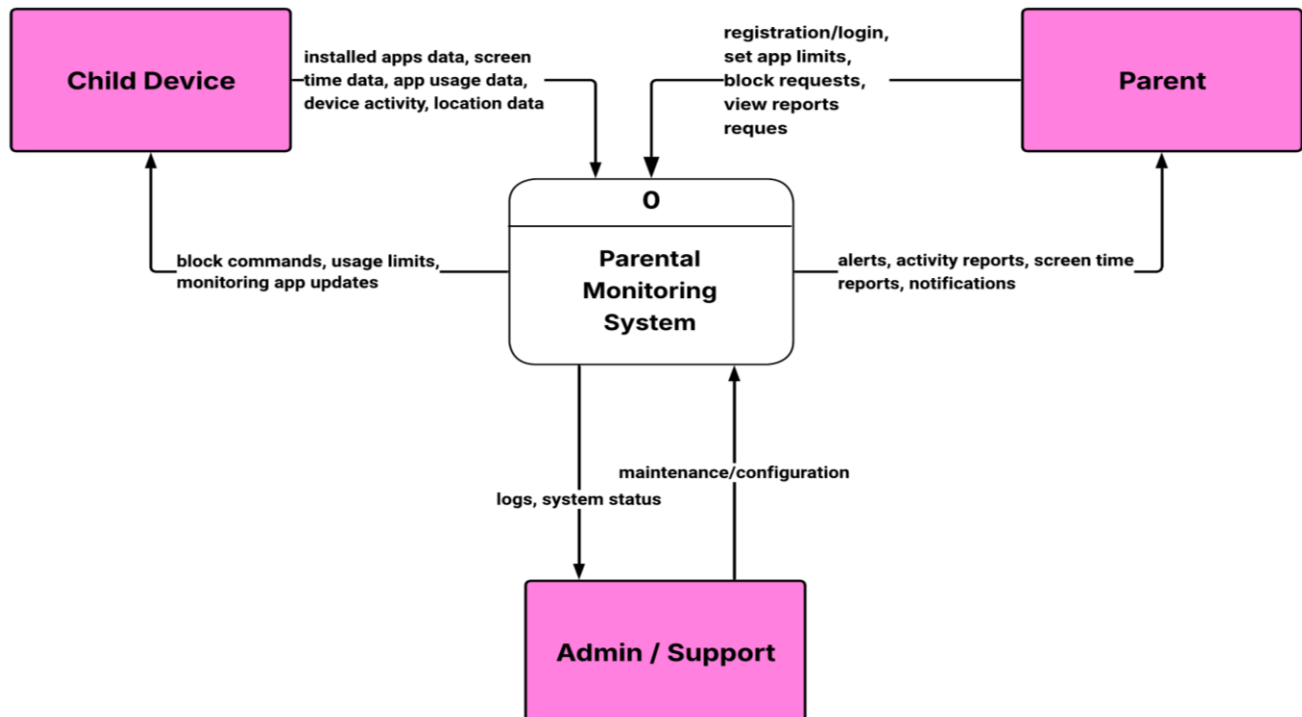
Break even Point = (initial investment) / (Net Cash flow) =
 (\$50,000) / (\$90,000 - \$35,000) = 0.91 Years

Which occur during the 11 Month from the first year.

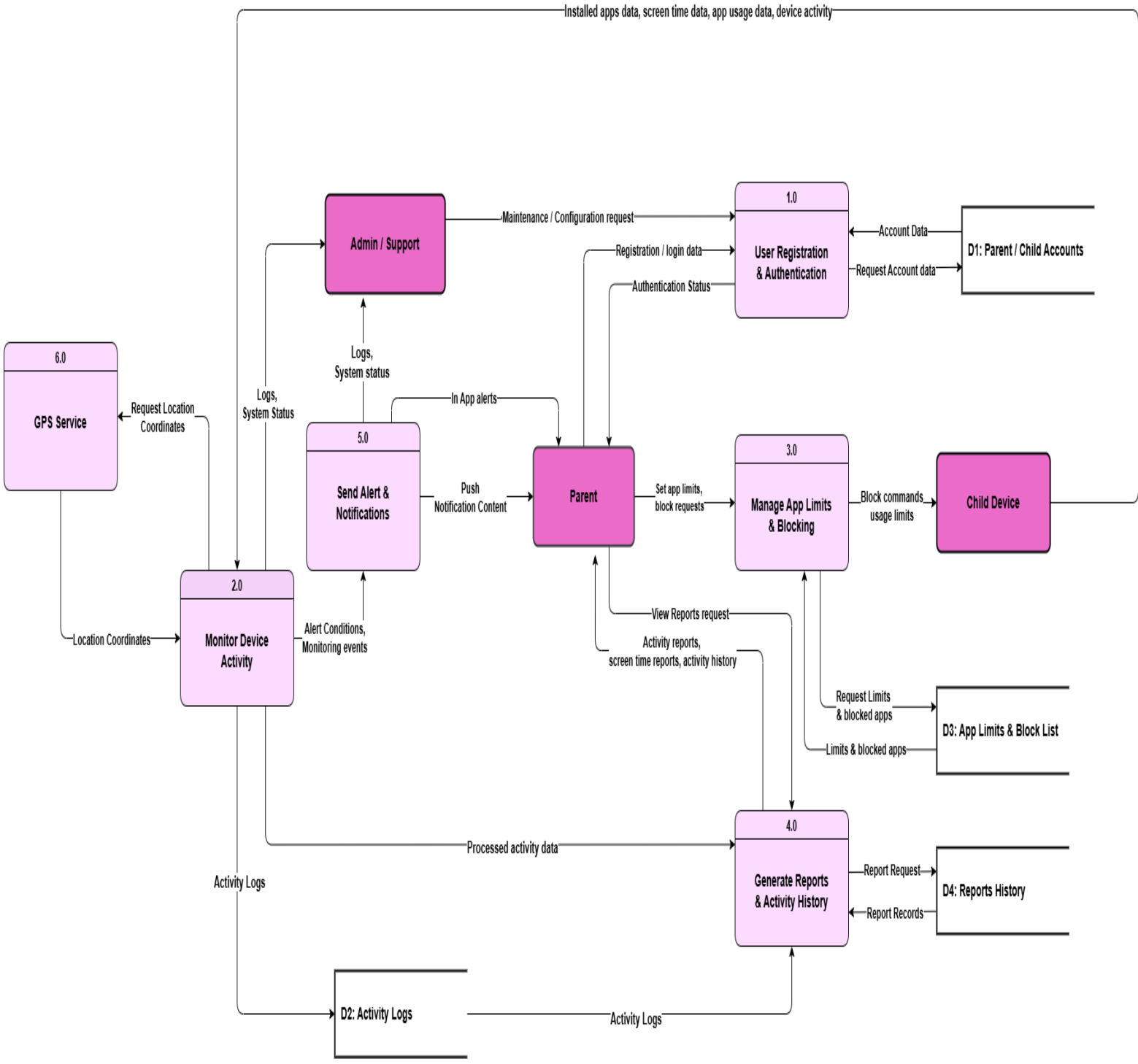


Section 4. Analysis

4.1 DFD Context :



4.2 DFD diagram :



4.3 Decision Table :

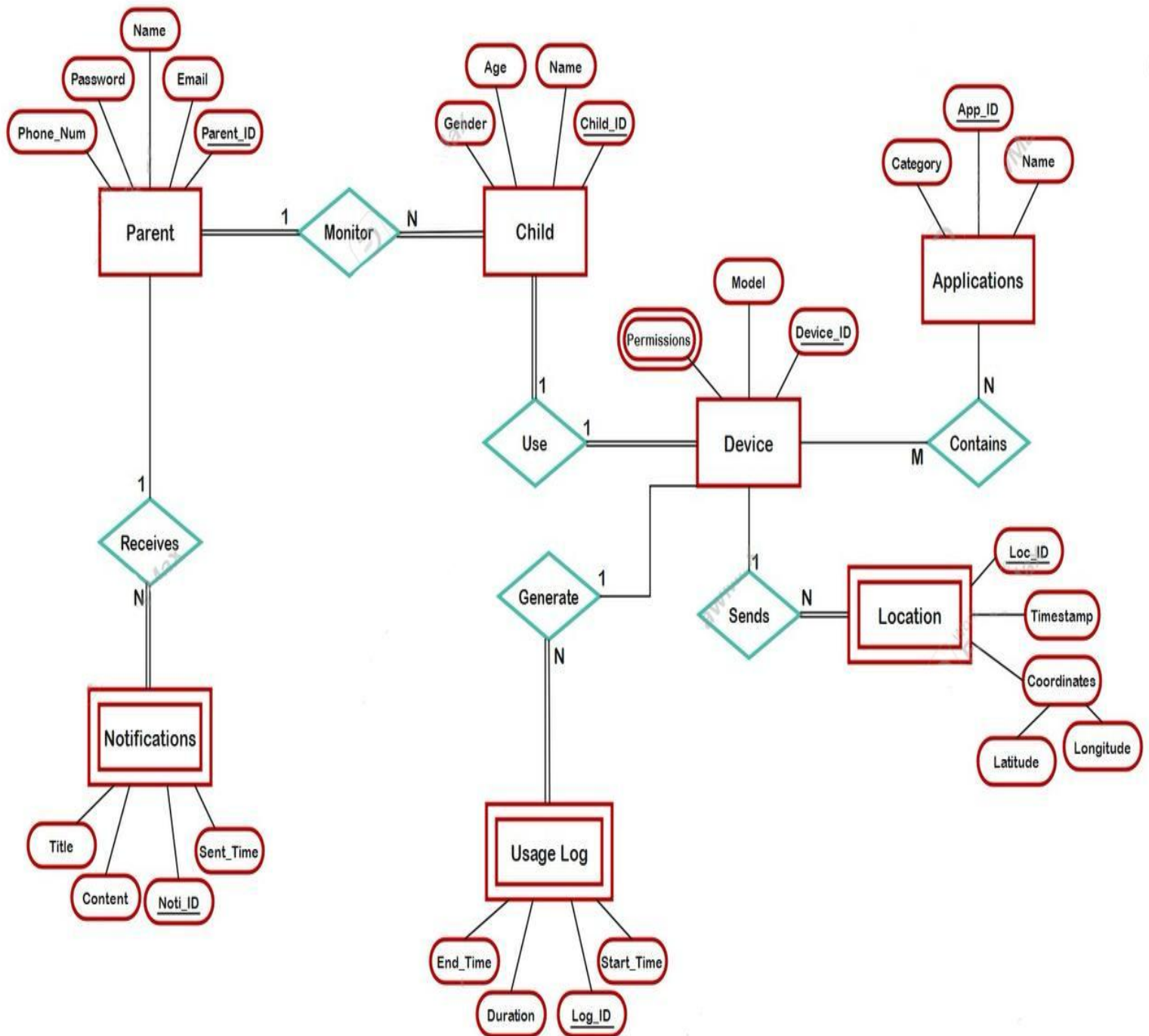
Conditions \ Courses of Action	Rules				
	R1	R2	R3	R4	R5
Block List Status	I	O	O	O	O
Usage Constrains	-	E	W	W	W
Location Safety	-	-	U	S	S
Content Appropriate	-	-	-	N	A
Allow Access					x
Instant Alert	x	x	x	x	x
Block Application	x	x	x	x	
Log Activity	x	x	x	x	x

Where :

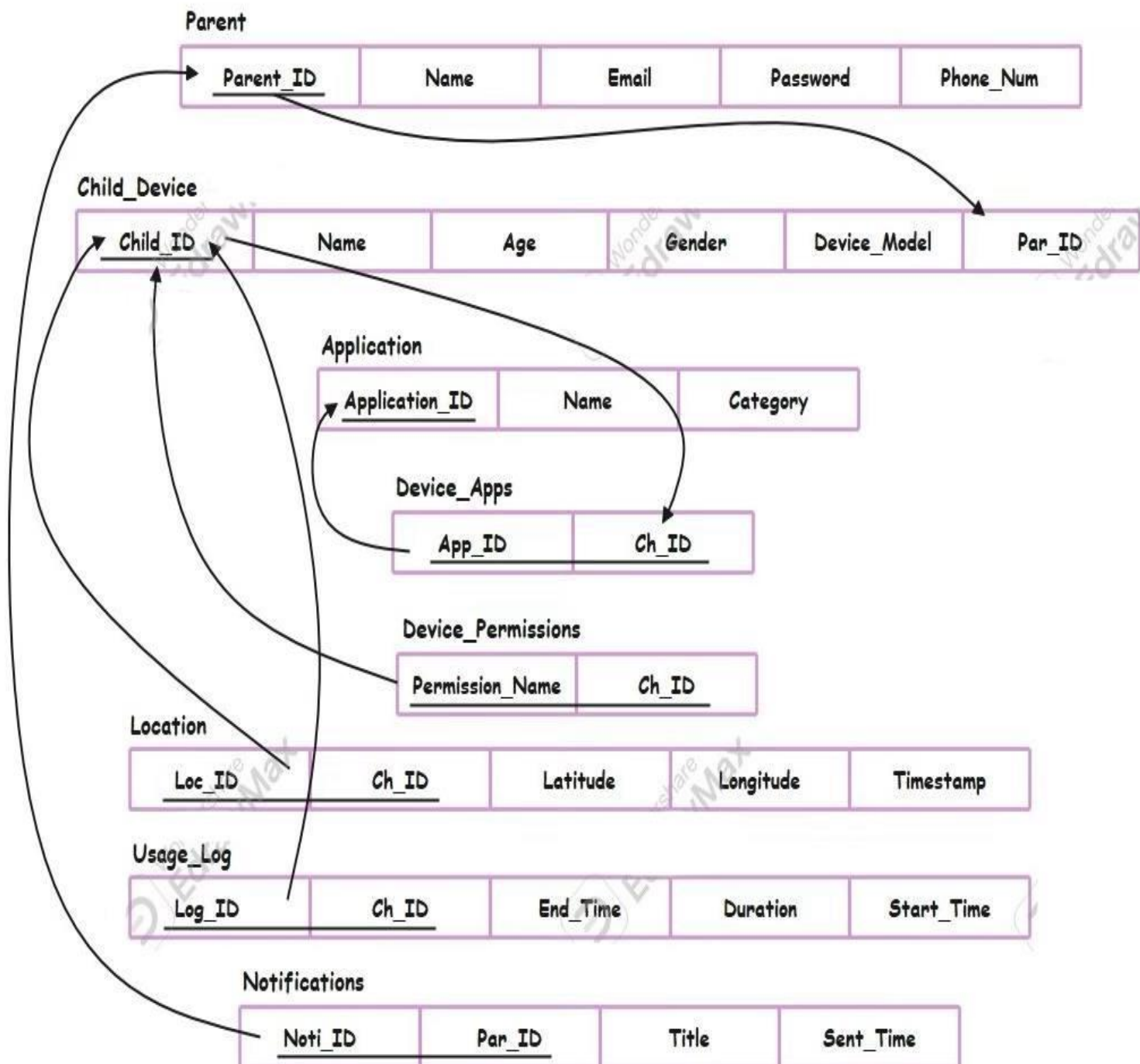
Block List Status	Usage Constrains	Location Safety	Content Appropriate
I: In list	E: Exceeded	S: Safe	A: Appropriate
N: Out of list	W: Within Limits	U: Unsafe	N: Not Appropriate

Section 5. Design

5.1 ERD:



5.2 Mapping:



5.3 GUI:



SafeChild



SafeChild

Empowering digital safety for your family

Welcome Back

Please enter your details to sign in

Email Address



name@example.com

Password

[Forgot Password?](#)



.....

Login

OR CONTINUE WITH



Google

iOS

Apple

Don't have an account? [Sign Up](#)

SafeChild

Alex

10 Years Old · iPhone 13

Online Now

82%

Screen Time

Daily Limit: 4h 00m

Details

2h 36m

Used Today

1h 24m

Remaining

-12%

vs Yesterday

Current Location

Lincoln Junior High

Last seen 4 mins ago

Most Used Apps

YouTube Kids

Entertainment

54m

Duolingo

Education

32m

Roblox

Games

25m

Lock Device

Play Sound

Pause Limits

Quick Message

Dashboard

Limits

Location

Reports

Alerts

SafeChild

TODAY'S USAGE

Leo's Screen Time

2h 45m of 4h limit

Leo is spending 20% less time than yesterday. Doing great!

Critical Alerts

Suspicious activity detected in "Messages" app 15m ago.

Review Alerts

Activity Peak

Last 7 Days

Mon Tue Wed Thu Fri Sat Sun

12 Apps Restricted

3 attempts blocked today

Manage Blacklist

Parent Dashboard

CURRENT LOCATION

Lincoln Junior High School

Digital Wellness Tip

Did you know? Setting a "No-Screen" hour before bedtime improves your child's sleep quality by 30%.

Try it now

Dashboard

Limits

Location

Reports

Alerts

SafeChild

Screen Time

Monitoring activity for Leo's iPhone

Daily Weekly

TOTAL 4h 12m

CATEGORY BREAKDOWN

Games 2h 15m

Social 1h 10m

Education 47m

Smart Insight

Leo has used gaming apps 15% more than the daily average. Consider setting a downtime window at 8:00 PM.

Set Downtime

App Breakdown

Sort by usage

YouTube

Entertainment

1h 45m

Limit: 2h 00m 15m left

Roblox

Gaming

58m

Limit: 2h 00m 1h 02m left

Duolingo

Education

45m

No Limit Set Goal Met

Instagram

Social

32m

Limit: 50m 18m left

Safari

Productivity

12m

No Limit Safe Search: ON

Add App Limit

Dashboard Limits Location Reports Alerts



SafeChild



CURRENT LOCATION



Home, 124 Park Avenue

2m ago



Leo is here



Location History



Dashboard



Limits



Location



Reports



Alerts

SafeChild

Limits

Manage your child's digital footprint by selecting which applications are accessible. Blocked apps will be restricted immediately on their device.

Search installed apps...

All Apps

Social Media

Games

YouTube

Entertainment

☒

Instagram

Social Media

☐

TikTok

Social Media

☒

Spotify

Entertainment

☐

Discord

Social Media

☒

Roblox

Games

☒

Outlook

Education

☐

WhatsApp

Communication

☐

Pinterest

Social Media

☒

Automatic Schedules

Set recurring periods where all non-essential apps are automatically blocked, such as during bedtime or study hours.

Configure Schedules

Smart Protection

Our AI automatically flags apps with age-inappropriate ratings or risky content reviews from our community.

Learn more →

Dashboard

Limits

Location

Reports

Alerts

SafeChild

Recent Alerts

Stay updated on your family's digital well-being.

CRITICAL

2 mins ago

Suspicious Activity

An unrecognized login attempt was blocked on Leo's tablet from a new location (London, UK).

Verify Device

Dismiss

WARNING

45 mins ago

Excessive Screen Time

Maya has reached 90% of her daily limit for social media apps. 15 minutes remaining.

GEOFENCE

1 hour ago

Entered Safe Zone

Leo has safely arrived at "Central Elementary School".

REPORT

3 hours ago

Weekly Summary Ready

The usage report for the week of Oct 12-19 is now available. Screen time is down 12% across all devices.

SYSTEM

Yesterday

Update Completed

SafeChild was successfully updated on Maya's iPhone 13. New content filters are now active.

Dashboard

Limits

Location

Reports

Alerts

SafeChild

Weekly Report

Leo's digital activity from Oct 12 – Oct 18

SCREEN TIME

TRENDS

4h 12m avg / day

-12% from last week

Mon

Tue

Wed

Thu

Fri

Sat

Sun

Sleep Hygiene

8h 45m

Goal: 9h

Focus / Study Time

2h 15m

Avg Daily

Most Used Apps

View All

Roblox

12h 30m

YouTube Kids

8h 15m

Messages

2h 45m

Smart Insights

Activity Spike Detected

Leo spent 45% more time on educational apps this Tuesday. Consider rewarding this progress!

Schedule Optimization

Moving the tablet 'Quiet Time' to 8:30 PM instead of 9:00 PM could improve Leo's sleep consistency.

Update Schedule

Dashboard

Limits

Location

Reports

Alerts



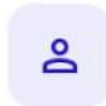
SafeChild



Settings

Manage your account and family security

ACCOUNT & DEVICES



Account Info

Update your name and email



Add Child Device

Sync a new phone or tablet



Privacy & Security

Password and two-factor auth



APP PREFERENCES



Notification Settings

Alert thresholds and methods



Support Center

FAQs and direct assistance



 Logout



Dashboard



Limits



Location



Reports



Alerts



SafeChild

Smart digital safety for the modern
family



SECURELY LOADING